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Scenario Based Quiz

Deep Learning, CNNs & Computer Vision

Scenario 1: Defect Detection in a Factory

A smartphone manufacturing company wants to **automatically detect scratches on phone screens** using images captured from a camera on the assembly line.

Q1.

The images captured are in **RGB format** and need to be processed before training the model. Which OpenCV function would you use to **read the image into Python**?

A. `cv2.open()`
B. `cv2.read()`
C. `cv2.imread()`
D. `cv2.load()`

`Answer: C. cv2.imread()`

Q2.

Before feeding images into the CNN, the engineer converts them to **grayscale** to reduce computational complexity.

Which OpenCV function is used for this transformation?

A. `cv2.resize()`
B. `cv2.cvtColor()`
C. `cv2.threshold()`
D. `cv2.blur()`

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Answer: B. cv2.cvtColor()
```

Q3.

To highlight scratches on the screen edges, the engineer decides to use **edge detection**.

Which algorithm is most commonly used?

A. Gaussian Blur
B. Canny Edge Detection
C. Median Filter
D. Morphological Closing

Answer: B. Canny Edge Detection

Scenario 2: CNN Feature Learning

A data scientist builds a **CNN model to classify images of cats and dogs**.

Q4.

The **convolution layer** in a CNN mainly performs which function?

A. Reduces the image size
B. Extracts spatial features such as edges and textures
C. Converts images to grayscale
D. Stores the training data

Answer: B. Extracts spatial features such as edges and textures

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Q5.

After convolution, the model applies **Max Pooling**.

What is the main purpose of pooling?

A. Increase the image size
B. Reduce dimensionality and retain important features
C. Increase the number of filters
D. Normalize the image pixels

Answer: B. Reduce dimensionality and retain important features

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Q6.

Which of the following represents a **typical CNN architecture order**?

A. Pooling → Fully Connected → Convolution
B. Convolution → Pooling → Fully Connected
C. Fully Connected → Convolution → Pooling
D. Pooling → Convolution → Softmax

Answer: B. Convolution → Pooling → Fully Connected

Scenario 3: Training an Image Classifier

A team is training a CNN model to classify **plant diseases from leaf images**.

Q7.

Which of the following is **commonly used as the loss function** for multi-class image classification?

A. Mean Squared Error
B. Binary Cross Entropy
C. Categorical Cross Entropy
D. Hinge Loss

Answer: C. Categorical Cross Entropy

Q8.

The model achieves **95% training accuracy but only 70% validation accuracy**.

This is an example of:

A. Underfitting
B. Overfitting
C. Perfect model
D. Data normalization

Answer: B. Overfitting

Q9.

Which technique helps **improve model accuracy and prevent overfitting**?

A. Increasing batch size only
B. Data augmentation
C. Removing convolution layers
D. Reducing dataset size

Answer: B. Data augmentation

Scenario 4: Transfer Learning

A startup wants to build an **animal recognition app** but has only **200 labeled images**.

Q10.

Which technique is most appropriate in this case?

A. Train CNN from scratch
B. Transfer learning using pretrained models
C. Use linear regression
D. Reduce image resolution

Answer: B. Transfer learning using pretrained models

Q11.

Which pretrained CNN model is commonly used for **transfer learning**?

A. K-Means
B. ResNet
C. PCA
D. Apriori

Answer: B. ResNet

Scenario 5: Real World Computer Vision Application

A city government installs cameras to **automatically detect traffic violations**.

Q12.

Which computer vision task is most relevant?

A. Image segmentation
B. Image classification / object detection
C. Speech recognition
D. Text summarization

Answer: B. Image classification / object detection

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